

System Design Specification

Engineering architecture for L1a, L1b, L2, L3 and technical handover.

Edition	V0.1 / Delivery Package V1.0
Status	Implementation baseline
Date	2026-08-21
Method	Workflow Atlas v2.1
Schema target	Dynamic N+3
Baseline route	Path B

Automatic outputs are research drafts. Researcher-reviewed TextGrid, transcript and matrix files are the final research data.

1. Purpose and evidence boundary

- Separate probabilistic AI evidence, deterministic transformation and researcher confirmation.
- Keep the original WAV as the canonical clock and preserve full-timeline alignment.
- Treat the three-speaker Multilogue04 Gold as calibration evidence, not a corpus-wide accuracy claim.
- Retain unsupported values as missing or flagged rather than manufacturing a numeric result.

Current implementation boundary

L1a provider artifacts and the L1b review path exist. Candidate listening/selection and dynamic N+3 are planned. L2/L3 contain feasibility scripts and prototypes, not a complete production workflow.

2. End-to-end architecture

Room-mix WAV	L1a evidence	Researcher mapping	L1b N+3 drafts	Praat review	L2 / L3
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Layer	Primary input	Processing responsibility	Primary output	Human gate
L1a	Room-mix WAV and run configuration	Diarization evidence, clips, candidate decisions and mapping	Accepted S1-SN handoff and speaker artifacts	Confirm candidates and mapping
L1b	Accepted L1a handoff and thresholds	Praat/acoustic timing, floor and nine labels	N+3 TextGrid drafts and reviewed metrics	Praat review
L2	Reviewed L1 data, transcript and signed definitions	Transcript units, lexical/MWU and pause/rate features	Feature tables and unresolved items	Definition/alignment review
L3	Accepted upstream outputs and codebook	Matrix merge, validation, report and archive	Final matrix, workbook and release package	Release acceptance

3. L1a: speaker evidence and candidate review

- Run audio preflight and provider diarization in Auto mode or with a fixed count prior.
- Normalize candidate turns to the WAV timeline and derive activity summaries.
- Generate representative playable clips from early, middle and late task regions.
- Let the researcher Include, Exclude, mark Uncertain or Merge candidates, then map accepted candidates to S1-SN.
- Rebuild RTTM/CSV, speaker TextGrid and full-duration muted-mirror WAVs from the accepted decision record.

Speaker-count constraint

A fixed count constrains clustering. It does not identify a teacher or guarantee that non-participant speech has been excluded. The candidate review gate remains required.

Target L1a API

Route	Responsibility	Status
POST /api/l1a/run	Create a provider/preflight run	Planned
GET /api/l1a/runs/:id/candidates	Candidate metrics and clip descriptors	Planned
GET /api/l1a/runs/:id/audio	Authorized range-capable clip stream	Planned
PUT /api/l1a/runs/:id/review	Save versioned draft decisions	Planned
POST /api/l1a/runs/:id/confirm	Seal mapping and rebuild Phase I artifacts	Planned

4. Dynamic N+3 interaction schema

For N accepted participants, the TextGrid contains N speaker tiers plus floor, transitions and flags. The three-speaker Gold therefore has six tiers. Current core code fixes S1/S2/S3 and requires a run-scoped refactor before broader N is an implemented claim.

Tier family	Count	Contents
Speaker	N	Exactly one of s, f, bc, ol, op, pf, tr, shs or x per interval
Floor	1	FREE or one canonical speaker
Transitions	1	Handoff point/status and Path B missing-overlap semantics
Flags	1	Uncertainty, conflicts and review evidence

5. L1b: acoustic timing and interaction labels

- Consume only a confirmed Phase I handoff.
- Run P025 and P035 independently; keep the threshold list configurable.
- Record full-timeline Scale Times behavior and the 200 s intensity window.
- Apply R1-R5 to floor state and use the controlled nine-label vocabulary.
- Use Path B when overlap is present: offset not measured, never zero.
- Upload Praat-reviewed TextGrids and recompute final duration metrics from reviewed boundaries.

Nine labels

Code	Meaning	Operational boundary
s	Speech	Lexical speech by the holder
f	Filled hesitation	Nonlexical holder hesitation; floor retained
bc	Backchannel	Short non-projecting listener response
ol	Overlap	Qualifying simultaneous speech
op	Own pause	Holder silence with retained floor
pf	Pause while floor held	Non-holder listening interval
tr	Transition gap	FREE no longer than L before another speaker
shs	Shared silence	FREE longer than L or to task end
x	Unusable/non-speech	Cough, handling noise or unusable event

6. L2 and L3

L2 starts only with reviewed L1 artifacts, a reviewed verbatim transcript and the research team's operational definitions. Reviewed word alignment is included when the selected pause/MWU analysis requires word-level timing. L3 merges accepted outputs only after the matrix schema and codebook are signed.

Layer	Required inputs	Output boundary
L2	Transcript conventions; AS-unit/clause rules; MWU definition; repair and rate rules; tool settings; Gold examples	RAW/TIDY, mappings and feature tables; unresolved values stay pending
L3	Accepted L1/L2 artifacts; field codebook; expected rows; validation rules	Final matrix/workbook, validation package and archive-ready export

7. WebUI, security and failure behavior

- One active job at a time with visible Layer state, artifacts and review handoff.
- Run-scoped storage outside the source tree; authorized path resolution for downloads and audio streams.
- Server-side API credentials only; no keys, signed URLs or transcript text in public logs.
- Timeline/schema failure blocks a research TextGrid; provider failure does not silently switch model.
- Upstream review changes invalidate dependent downstream outputs.

8. Validation and acceptance

Layer	Minimum evidence
L1a	Provider contract tests, artifact invariants, review persistence and mapping audit
L1b	Timeline/schema invariants, deterministic replay, Gold metrics and Praat review finalization
L2	Definition fixtures, transformation logs, expected feature tables and alignment evidence where used
L3	Matrix schema reconciliation, representative rows, browser QA and archive verification
Publication claim boundary Multilogue04 is calibration evidence. A separate researcher-corrected multilogue is required before claiming generalization of the frozen method.	

End of specification

The Workflow Atlas remains the research-facing method reference. This specification is the engineering implementation companion and must be updated when the signed method, schema or accepted release changes.